

Project concept note

Crabbs Seawater Reverse-Osmosis Desalination Plant

Purpose

Add 15,000 m³/day of drought-proof potable water for Antigua, replacing ageing diesel-driven units and reducing non-revenue water pressure on the public utility.

Location

Crabbs Peninsula industrial area, adjacent to existing utility land and the north-coast distribution main.

Scope

Open seawater intake, pre-treatment (DAF + UF), two-pass reverse osmosis, remineralisation, 20,000 m³ product storage, 6 MWp solar with battery support, brine diffuser outfall.

Structure

Design-build-operate-transfer PPP, 25-year term, with the public utility as sole offtaker.

Indicative capital cost

US\$48 million (illustrative).